

Chapter 15 — Lower Urinary Tract Catheterisation

HISTORY

Era	Development
1500 BCE	Earliest record: Egyptian transurethral bronze tubes , reeds, straws and curled palm leaves
400 BCE	Hippocrates: malleable lead tubes
CE 79	S-shaped silver tube for urinary retention found at Pompeii
4th–5th century	Byzantine anthology lists indications (fistula repair, retention, kidney "inflammation"); Latin authors describe catheterising paralysed patients
Middle Ages; 16th century	Use recorded in Spain and China; indwelling wax-and-cloth catheter documented
18th century	Benjamin Franklin's silver wire catheter for his brother's retention from a bladder calculus; later used on himself
1855	Reybard : first self-retaining catheter — two channels, one draining urine, one inflating a balloon
1929	Foley : modern rubber catheter with inflatable balloon — designed for post-prostatectomy haemostasis , then widely adopted
1968	Silicone catheters : less urethritis, encrustation and infection
1971	Lapides : clean intermittent catheterisation (CIC) — infection arises from stasis and overdistension , not bacteria alone; initially scorned, now widely accepted

Concept — the two modern advances are the **self-retaining balloon (Reybard → Foley)** and **CIC (Lapides)**; the second rests on the insight that **regular emptying, not sterility, prevents infection**.

ANATOMIC CONSIDERATIONS

Male Urethra

- Length **18–20 cm**. The International Consultation on Urological Diseases (ICUD, Morocco, 2010) divides it into **seven areas**: meatus, fossa navicularis, penile, bulbar, membranous, prostatic urethra, and bladder neck (Fig. 15.1 numbers the six beyond the meatus).
- By common usage the **anterior urethra** = fossa navicularis, penile (pendulous) and bulbar; the **posterior urethra** = membranous, prostatic and bladder neck.

Female Urethra

- About **4 cm**, lying within the **anterior vaginal wall**.

NORMAL URETHRAL CALIBRE

Site	Accommodates
Male meatus	24 Fr
Prostatic urethra	32 Fr
Bladder neck	28 Fr
Female urethra	22 Fr

- The trap is the order: the **meatus is the narrowest point of the male urethra** (24 Fr), narrower than bladder neck (28 Fr) and prostatic urethra (32 Fr).

INDICATIONS FOR LOWER URINARY TRACT DRAINAGE

- Drainage is either **indwelling (Foley)** or **straight (intermittent)**. Straight catheterisation serves **temporary drainage in acute retention** and **sterile urine collection**, especially in children and women.

TABLE 15.1 LIST OF COMMON INDICATIONS FOR PLACEMENT OF AN INDWELLING URETHRAL CATHETER

Purpose	Indications
Relieve retention or poor emptying	Acute retention; chronic retention; incomplete emptying; urosepsis with incomplete emptying
Bleeding	Gross haematuria and clot retention
Protect against high pressure	High-pressure voiding; high-pressure (poorly compliant) bladder
Care and monitoring	Immobility; lengthy surgery; strict output monitoring
Postoperative healing	After urologic surgery, e.g. urethroplasty, prostatectomy

- The list reduces to four aims — **empty, protect the upper tracts, monitor, allow healing**.

Contraindications

- **Absolute: known or suspected urethral injury.** Suspect it with **pelvic fracture**, gross haematuria, **blood at the meatus**, perineal ecchymosis, and a **high-riding prostate** on digital rectal examination (DRE) → **retrograde urethrogram (RUG) before any Foley**. Blind passage is avoided except in exceptional emergencies — then with a well-lubricated catheter by an experienced team member.
- **Relative: an artificial urinary sphincter (AUS)**, which must be deactivated first; **urethral stricture**, which may need additional manoeuvres.

TECHNIQUE OF URETHRAL CATHETERISATION

Preparation for Catheterisation

- Standard tray: sterile gloves, catheter, sterile lubricant, antiseptic, drapes, sterile water, collection bag with tubing. **Specialty catheters and wires are not in basic kits** — have them ready according to the indication.
- Sequence: open tray sterilely, glove → (optionally test the balloon with sterile water — not routine) → antiseptic onto swabs → sterile field with drapes → prepare the genitalia.

Urethral Catheterisation in Men

- Supine → retract foreskin, sterilise glans → **inject 10–15 mL lubricant into the meatus**, pinch glans to prevent reflux → the non-dominant hand is now non-sterile → dominant hand inserts with the **penis straight and taut** → slight resistance at sphincter or prostate is normal; slow, steady pressure → **advance to the hub** (balloon port at the meatus) → confirm urine flow → **inflate with sterile water** → withdraw to seat the balloon → **replace the foreskin** (prevents paraphimosis).
- **Urine may return before the bladder neck is passed:** inflating then places the balloon in the prostatic urethra — insert fully **and** confirm flow before inflation.
- Beyond the bladder neck with no urine → lumen **plugged by lubricant** → irrigate with sterile water or saline.
- Adjuncts: **2% lidocaine gel** for apprehensive patients or failed attempts; **distraction** to pass the external sphincter.

Urethral Catheterisation in Women

- Usually easier — **shorter urethra**. Supine **frog-leg** → thumb, index and middle fingers of the non-dominant hand spread the labia minora, **lifting anteriorly** → sterilise meatus → insert lubricated catheter; little or no resistance → **inflate once urine returns**.
- Poor view → **Trendelenburg** and assistant retraction.
- Examinable pair: **men — inflate only at the hub after urine; women — inflate on urine return**.

Urethral Catheterisation of Children

- Same principles. **Physiological phimosis must not be forcibly retracted** — move the foreskin ventrally or dorsally to find the meatus.
- Boys may show **intense sphincter contraction**: hold steady pressure without force until it releases, often with a deep breath. Size by age (Table 15.2).

TABLE 15.2 RECOMMENDED SIZE OF PEDIATRIC CATHETERS ACCORDING TO AGE

Age	Straight (Fr)	Balloon (Fr)
Newborn	4–6	6
1–11 months	6–8	6–8
12–23 months	8	8
2–6 years	10	8–10
7–12 years	10–12	12–14
Over 12 years	14	14–16

- Size climbs from about 6 Fr at birth to adult sizes after 12; **8 Fr covers the second year**, and the balloon catheter runs slightly larger than the straight one except at 2–6 years.

Types of Urethral Catheters

Catheter Sizing

- Choose by **patient factors, anatomy and indication** — a young man in retention after hernia repair needs a different catheter from an elderly man in clot retention after transurethral resection of bladder tumour (TURBT).
- Size is in **French (Fr)**; French size and recommended **balloon volume** are printed on the balloon-port cuff.
Diameter (mm) = Fr ÷ 3; larger catheters are also **more rigid**.

TABLE 15.3 APPROXIMATE CONVERSION FROM FRENCH TO DIAMETER IN MILLIMETERS (ANCHOR VALUES)

French	3	6	12	16	18	20	24	30	33
Diameter (mm)	1	2	4	5.33	6	6.67	8	10	11

Values approximate; every intermediate size follows the same Fr ÷ 3 rule (table runs 3–33 Fr).

Catheter Types

Tip	Feature	Use
Straight	Standard; supplied in kits	Routine
Coudé ("elbow")	Curved tip; knob near the balloon port marks tip orientation	Prostatic enlargement — follows the upward bulbar-prostatic curve; tip pointed to the ceiling (patient supine)
Councill	Open distal tip (standard is blind-ending)	Passed over a wire

Lumens	Ports	Use and limits
One	Drainage only	

Lumens	Ports	Use and limits
		Straight catheterisation: temporary relief, specimen; cannot be anchored
Two	Drainage + balloon	Most common ; drains and anchors
Three ("haematuria catheter") (Fig. 15.2)	Drainage + balloon + irrigation inflow	Continuous bladder irrigation after clot evacuation; larger drainage eyes ; rigid wall or metal tip spring resists collapse during hand irrigation

- **Each added port shrinks the drainage lumen**: a single-lumen 20 Fr drains better than a three-way 20 Fr.

Materials and Coatings

- Materials: **latex, silicone, rubber, polyvinyl chloride (PVC)**; coatings: **antibiotic, hydrophilic, antiseptic**.

Comparison	Asymptomatic bacteriuria	Symptomatic urinary tract infection (UTI)
Silver oxide vs standard	Pooled trials: insufficient evidence to recommend	
Silver alloy vs standard	Reduced ($>10^5$ colony-forming units)	No difference when used as primary endpoint
Minocycline–rifampicin vs standard (after prostatectomy, small trial)	Reduced	No difference
Silicone, latex, hydrophilic-coated latex, PVC compared	No difference	—

- **Silicone** gave fewer reported urethral symptoms than non-silicone catheters, and less swab-diagnosed urethritis than latex.

Concept — coatings change **bacterial counts, not clinical infection**; silicone's genuine advantage is **urethral tolerance**.

Diagnostic Catheters

- **Lapides diagnostic Foley**: **five radiopaque rings 1 cm apart** — calibrates **female urethral length**.
- **Davis and Trattner** catheters: **two balloons** (bladder neck + meatal seal) so contrast can be instilled into the urethra — diagnosed **female urethral diverticula**; **largely replaced by magnetic resonance imaging (MRI)** (Fig. 15.3).
- **Urodynamic catheter**: three ports — **pressure sensor, filling, balloon**.

Drain Catheters

- **Malecot** (Fig. 15.4): single lumen, **no balloon**; **four-winged tip** compressed for insertion, holds in place. Used postoperatively as **nephrostomy or suprapubic (SP) tube**, or placed into a **conduit after continent diversion**.

Condom Catheters

- **External drainage** for men with **limited mobility and no outlet obstruction**.

KEY POINTS: CATHETERS

- There are many indications for urethral catheterisation and many sizes and types of catheter.
- As French size increases, the catheter lumen diameter increases (Table 15.3).
- Three-way haematuria catheters should be used if clot retention is suspected.

SUPRAPUBIC CATHETERISATION

- SP tubes **bypass the urethra**, draining through the lower anterior abdominal wall; placed at the bedside or by minor surgery.
- **Indications: urethra inaccessible** (disruption from trauma; acute retention with significant stricture); **permanent catheterisation**, where SP may be more comfortable.
- **Advantages: easy, comfortable exchange** (immobility, contractures, difficult genital anatomy); avoids penile trauma, stricture, prostatic bleeding, **traumatic hypospadias/urethral erosion**, a **patulous female urethra**; less interference with sexual activity.
- **Complications:** bowel injury or perforation; abdominal wall cellulitis or abscess.
- **First exchange at 4–6 weeks** by experienced staff — a new cystotomy tract is easily lost; later exchanges can be done by almost anyone once the tract matures.

Techniques

- **Open vs percutaneous: percutaneous preferred** — bedside, local anaesthesia.
- Contraindications to percutaneous placement (absolute and relative together): **prior abdominal or pelvic surgery violating the retropubic space**, complex anatomy, **bowel overlying the bladder**, no bedside ultrasound, surgeon preference.

Minimally Invasive

- Two kit types — **trocar-based** and **self-contained Seldinger** — same principles; **ultrasound guidance** for both. Sequence:
 1. Ultrasound confirms a **full bladder with no overlying bowel**.
 2. **Trendelenburg** — moves peritoneal contents cephalad.
 3. **1–2 cm horizontal midline incision, 1–2 fingerbreadths above the symphysis**.
 4. Spinal needle into the bladder, aspirate urine; infiltrate local anaesthetic along the tract; confirm by ultrasound or cystoscopy.
 5. (a) **Trocar:** trocar and sheath (stainless-steel obturator and outer sheath, Fig. 15.5) into the bladder → urine returns → remove trocar → **16-Fr catheter through the sheath** → remove sheath. (b) **Seldinger:** wire through needle → gentle tract dilation → catheter.
 6. Do not advance the catheter too far — it may enter the bladder neck or urethra.

Open Cystotomy

- General anaesthesia or sedation → **distend bladder to capacity** → small midline or Pfannenstiel incision → expose and incise rectus fascia, spread rectus → palpate bladder, **stay stitches** lift it → small cystotomy → any catheter placed and anchored → haemostasis → secure to skin with permanent suture.
- **Lowsley retractor** (adapted urethral sound): advanced per urethra into bladder → pushed anteriorly to the abdominal wall → skin incision, cautery down onto its tip → tip exposed → catheter tied to it (2-0 silk) → retractor and catheter pulled back out through the urethra and meatus → tie cut → catheter guided cystoscopically back into the bladder → balloon inflated → secured to skin with permanent suture.
- Pair: **percutaneous — bedside, local, ultrasound-guided; open — theatre, anaesthesia, direct vision**, reserved for when percutaneous access is unsafe.

DIFFICULT CATHETERISATION

Prostatic Obstruction

- **Benign prostatic hyperplasia (BPH) is the commonest cause:** adenoma narrows the prostatic urethra; with the male urethral angle this blocks advance or creates an **iatrogenic false passage posterior to the prostatic urethra**.
- **14–16 Fr usually will not pass; 18 Fr is more rigid** with more tip surface area. **Suspected BPH → at least 18-Fr coudé**, marker kept **anterior** so the tip points anteriorly — usually sufficient.
- **False passage:** suspect with **blood in the port or on the tip** after a difficult pass → **cystoscopy** rather than further blind attempts (which extend the passage) → traverse it visually, usually angling the camera anteriorly → guidewire into bladder → **Councill catheter over the wire**.

Urethral Stricture

- Strictures occur anywhere; **location estimated by the length of catheter passed before resistance**. Gentle pressure may dilate a soft stricture.
- **Membranous strictures are most prone to false passage** (urethral angle), though it can happen anywhere.
- Sequence: **atraumatic hydrophilic guidewire** past the stricture → **confirm in bladder by passing a 5-Fr catheter and aspirating urine** (or place under vision with flexible cystoscopy) → exchange for a **stiffer wire through the 5-Fr catheter** → **stepwise dilation**, small to large, not too deep (posterior bladder wall perforation) → **dilate one size above the planned catheter** (16-Fr catheter → dilate to at least 18 Fr), since the catheter is less rigid than the dilator → **Councill catheter over the wire**.
- Alternatives: balloon dilator, filiform dilators. Dense stricture or intolerance → theatre, or **SP catheter**.

Urethral Trauma

- Suspected injury → **RUG before catheterisation**; blind passage avoided except in exceptional circumstances. **Pelvic fracture urethral injury** → **SP tube is the preferred initial management**.

FIG. 15.6 ALGORITHM FOR DIFFICULT CATHETER PLACEMENT

- **First attempt fails — prostatic obstruction suspected** → 18-Fr coudé → fails → cystoscopy → *stricture*: guidewire, dilate, Councill over wire; *false passage*: super-stiff wire exchanged over 5-Fr catheter, Councill over wire.
- **Not suspected** → smaller-diameter catheter → fails → guidewire → if placed: stricture → dilate, Councill over wire; false passage → Councill over wire → if guidewire fails → cystoscopy, same branches.
- Footnotes: **wire cannot be placed** → **SP tube**; SP also for urethral disruption or undilatable dense stricture; dilation is not appropriate for every stricture.

Abnormal Anatomy

Posterior Urethral Valves

- Newborn boys need **urgent catheterisation**. The catheter **coils because of the high bladder neck**. Try **6–8 Fr**; premature urethras may be too small → **5-Fr open-ended catheter over a guidewire** → failure → **SP tube or vesicostomy**.

Obesity

- The problem is **exposure** — most difficult female catheterisations are failures of exposure. Trendelenburg; assistants (and the awake patient) retract the suprapubic fat pad or pannus; headlamp. Female meatus still hidden → **two fingers in the vagina, pass the catheter over them guiding anteriorly**.

- **Buried penis:** infrapubic manual pressure reveals the meatus; dense cicatrix → **flexible cystoscopy to find it. High incidence of urethral stricture** in buried penis — expect adjunctive manoeuvres.

Reconstructive Considerations

- Full history, examination, imaging and operative notes before instrumenting a complex patient; after **bladder neck closure** urethral catheterisation is impossible.
- **Neobladder:** usually catheterisable per urethra unless anastomotic stenosis or tumour recurrence. If not, **SP placement is hazardous — the neobladder is intraperitoneal** (bowel or pedicle injury) → **interventional radiology-guided or open SP.**
- **Continent reservoir:** not catheterisable per urethra; reservoir usually right lower quadrant, channel there or at the umbilicus. Commonest failure: **overdistension kinks the channel. Never force** (perforation, false passage) → flexible cystoscope, ureterscope or hysteroscope into the reservoir → wire → Councill; failure → SP with neobladder precautions.

Artificial Urinary Sphincter

- Three-piece hydraulic device for male stress incontinence; catheterisation is a **relative contraindication — cuff erosion**. If essential: **fully deactivate** → cuff size unknown → **smallest catheter, 12–14 Fr** → **remove as soon as possible**; prolonged drainage → **SP tube**.

KEY POINTS

- The urologist should be prepared to attain lower urinary drainage in almost all situations.
- Review the history and prior operative reports before instrumenting the lower urinary tract.
- If BPH is expected, use an 18-Fr coudé catheter.
- Avoid catheters with an AUS; if necessary, place a 12–14 Fr after deactivation and remove it as soon as possible.

Concept — every difficult catheterisation escalates the same way: **blind** → **wire** → **vision** → **suprapubic**; repeated forceful blind passage is the one step never taken.

COMPLICATIONS OF LOWER URINARY TRACT DRAINAGE

Infection

- Per the Centers for Disease Control and Prevention (CDC): **UTIs are the commonest healthcare-associated infection**, and **75% of hospital-acquired UTIs are catheter-associated**. Evidence-based guidelines on catheter-associated UTI (CAUTI) prevention and treatment: CDC and Infectious Diseases Society of America (IDSA).

Iatrogenic Trauma

- Usually from **poor technique or aggressive attempts in the difficult patient** → false passage, haematuria, **AUS cuff injury or erosion**; long-term indwelling → **traumatic hypospadias, urethral erosion**.
- About **0.3% of hospitalised patients** (one institution: about 3 per 1000 male admissions), usually presenting with **penile bleeding**. Occurs at **insertion, balloon inflation, or removal with an incompletely deflated balloon**.
- **Catheter safety training significantly reduced injuries** — the injury is preventable, and **education is the most important intervention**.

Malignancy

- Data from **spinal cord injury (SCI)**: higher bladder cancer risk, but overall only about **0.3%**, even with indwelling catheters. Cystoscopy and cytology are **poor screening tests** here (biopsies mostly inflammation and metaplasia; unnecessary biopsies) → the **American Urological Association (AUA)** **does not recommend screening or surveillance cystoscopy**.

The Retained Catheter

Cause	Management
Knotted catheter — over-advanced in a child or woman	Cystoscopy alongside the catheter; puncture balloon with a cystoscopic needle
Balloon will not deflate on aspiration	Stiff end of a wire through the balloon channel to puncture → cystoscopy to retrieve all balloon fragments
Calcified catheter — missed exchanges	Cystoscopic laser lithotripsy , then extraction

KEY POINTS

- Catheter-associated UTIs are the most common hospital-acquired infection.
- Catheters may lead to trauma, stricture and erosion.
- Chronic catheterisation can cause bladder inflammation, but routine surveillance cystoscopy is not recommended.